GROUP CASE 1

Comprehensive Financial Regression Analysis

Financial Regression and Time-Series Analysis | FINARTS C01

Group Assessment | 20% of Final Grade | Due: Week 7

Coverage	CLO1–CLO4 / Regression block
Software	R / RStudio / R Markdown
Assessment	Group Case 1 (20%)
Release	Week 2; progressive preparation through Week 6

Important: You are not expected to know every technique when the case is released. The case is designed to be completed progressively as the relevant regression methods are taught in Weeks 2–6.

1. Case Purpose

Your group has been engaged as a junior financial analytics team to investigate a financial decision using empirical data. Your task is not simply to estimate a regression model. You must develop a complete and defensible financial regression analysis: define the problem, prepare the data, specify and estimate a model, conduct statistical inference, evaluate and diagnose the model, improve it where necessary, apply an appropriate advanced regression technique, and translate the results into a financial recommendation.

2. Learning Outcomes Assessed

CLO	What your group must demonstrate
CLO1	Explain the principles, assumptions, and limitations of classical linear regression and its role in financial/economic analysis.
CLO2	Estimate, interpret, and evaluate multiple linear regression models, including coefficients, inference, goodness of fit, and financial meaning.
CLO3	Diagnose and address regression problems, including multicollinearity, heteroskedasticity, autocorrelation, outliers/influential observations, and model specification issues.
CLO4	Apply appropriate advanced regression methods such as functional forms, interactions, categorical variables, robust estimation, regularization, logistic regression, and/or panel-data methods.

3. Case Scenario

The instructor will provide your group with a financial/economic dataset and a decision problem. Your group must formulate a focused empirical question that can be investigated using multiple regression. The specific variables, sample period, and business context will be provided in the case dataset/brief or announced in class.

Your final analysis should answer a question of the following general form:

"Which financial or economic factors are associated with the outcome of interest, how strong are those relationships, and what should a decision-maker do based on the evidence?"

4. Required Analysis

4.1 Data Understanding and Preparation

- Identify the unit of observation and describe the dependent and explanatory variables.
- Check data types, missing observations, and obvious data-quality issues.
- Produce appropriate descriptive statistics.
- Create relevant visualizations and briefly interpret the important patterns.
- Explain any data transformations or exclusions made and justify them.

4.2 Initial Multiple Linear Regression Model

- Formulate and estimate an initial multiple linear regression model in R.
- Report the estimated coefficients, standard errors, t-statistics, p-values, confidence intervals, R-squared, adjusted R-squared, and the overall F-test.
- Interpret the coefficients in context using the ceteris paribus principle.
- Comment on whether the signs and magnitudes are economically plausible.

4.3 Statistical Inference

- Test relevant individual coefficients using appropriate null and alternative hypotheses.
- Use confidence intervals to assess parameter uncertainty.
- Distinguish statistical significance from practical/economic significance.
- Where relevant, use joint hypothesis tests and explain the result in financial terms.

4.4 Model Evaluation and Comparison

- Evaluate model fit using appropriate measures.
- Compare relevant alternative specifications where justified.
- Use adjusted R-squared, AIC/BIC, nested-model tests, or other methods covered in class where appropriate.
- Do not select a model solely because it has the largest R-squared.

4.5 Regression Diagnostics

- Assess multicollinearity using appropriate evidence such as correlations and VIF.
- Assess heteroskedasticity using appropriate diagnostic evidence.
- Assess autocorrelation when it is relevant to the structure of the data.
- Investigate outliers and influential observations.
- Use residual diagnostics to evaluate whether the model raises important concerns.

4.6 Model Improvement

- Identify the most important model problems found in the diagnostics.
- Apply appropriate corrective methods such as robust standard errors, weighted least squares, transformation, or model respecification where justified.
- Explain why the selected remedy is appropriate.
- Compare the improved model with the initial model and discuss the implications.

4.7 Advanced Regression Component

- Apply at least one advanced regression method that is genuinely appropriate to the financial problem and justified by the data or research question.
- Possible methods include functional forms, categorical variables, interaction effects, robust estimation, regularization, logistic regression, or panel-data methods.
- Do not force every advanced technique into the analysis. Methodological judgment is part of the assessment.

4.8 Financial Interpretation and Recommendation

- Summarize the main empirical findings.
- Explain the financial/economic meaning of the results.
- Discuss important limitations and remaining uncertainty.
- Provide a clear evidence-based financial recommendation directly linked to the analysis.

5. Central Principle of the Case

Do not use every technique simply because it is available. The quality of the case depends on your ability to identify the financial problem, select appropriate methods, justify those choices, and interpret the evidence responsibly. A technically complicated model is not automatically a better model.

6. Required Deliverables

Component	Requirement
1. R Markdown Report	Complete written analysis with financial context, methods, results, interpretation, and recommendation.
2. R Markdown Source (.Rmd)	The actual source file used to generate the report.
3. Supporting Files	Relevant data files and supporting R files needed to reproduce the analysis.
4. Rendered Report	HTML or PDF produced from the R Markdown workflow.
5. Group Recording	Submit a short group video (recommended 8–12 minutes) discussing the final regression output. The recording should explain the key findings, important diagnostics and remedies, the selected final model, and the financial recommendation. Group members should participate and be prepared to defend the group's methodological choices.

The submission should follow the reproducible workflow introduced in Module 1: Data → R Code → Results → Interpretation → Report. In addition to the written and reproducible files, your group must submit a short recording discussing and defending the final output.

7. Recommended Report Structure

Section	Suggested Content
1. Executive Summary	Financial problem, key findings, and recommendation.
2. Introduction	Decision context and empirical research question.
3. Data and Variables	Dataset, unit of observation, variable definitions, preparation, descriptive statistics.
4. Initial Regression Model	Model specification, estimation, coefficient interpretation, inference, and fit.
5. Diagnostics	Multicollinearity, heteroskedasticity, autocorrelation where relevant, influence/outliers, residual diagnostics.
6. Model Improvement	Problems identified, remedies applied, and comparison of results.
7. Advanced Regression	Appropriate advanced method, justification, estimation, and interpretation.
8. Financial Interpretation	Meaning of the results, limitations, and decision implications.

9. Recommendation and Conclusion	Final evidence-based recommendation and concise conclusion.

8. Group Work Expectations

- All group members are expected to contribute meaningfully to the analysis and final submission.
- The group should maintain a clear and reproducible file structure so that another member can trace data, code, results, and interpretation.
- Use group consultation opportunities during Weeks 2–6 to clarify methodology and troubleshoot the analysis.
- The final report should represent the group's own analysis and reasoning.

9. Suggested Timeline

Week	Group Action
Week 2	Receive case brief; understand the financial problem; inspect the dataset; begin variables and research question.
Week 3	Complete initial multiple regression, inference, and interpretation.
Week 4	Conduct diagnostics and identify model problems.
Week 5	Work on functional forms, categorical variables, interactions, and other relevant specifications.
Week 6	Complete advanced regression component, finalize diagnostics/remedies, prepare final interpretation, and check reproducibility.
Week 7	Submit Group Case 1, submit the group recording, and complete the assigned case activity/presentation format.

10. Assessment Rubric -

The case is scored out of 100 points and contributes 20% to the final course grade.

Criterion	Weight	What is assessed
1. Financial Problem, Research Question & Model Specification	10%	Clarity and financial relevance of the problem; justification of variables; appropriateness of the model specification.
2. Data Preparation & Descriptive Analysis	10%	Correct preparation; variable understanding; missing/data-quality checks; descriptive statistics; relevant visualizations and interpretation.
3. Multiple Regression Estimation & Interpretation	15%	Correct estimation; coefficient interpretation; ceteris paribus reasoning; model fit; financial meaning.
4. Statistical Inference & Model Evaluation	10%	t-tests, confidence intervals, F-tests, appropriate model-comparison evidence; distinction between statistical and economic significance.
5. Regression Diagnostics	15%	Appropriate assessment of multicollinearity, heteroskedasticity, autocorrelation where relevant, influence/outliers, and residual behavior.
6. Model Improvement / Corrective Methods	10%	Identification of problems; appropriate remedies; justification; comparison with the initial model.
7. Appropriate Advanced Regression Method	10%	Appropriate selection, implementation, justification, and interpretation of an advanced method.
8. Financial Interpretation & Recommendation	10%	Quality of financial/economic interpretation, limitations, and evidence-based recommendation.
9. Reproducibility, R Implementation & Professional Presentation	5%	Traceable R workflow, reproducible files, professional tables/figures, organization, and presentation quality.
10. Group Recording Results Discussion & Defense	5%	Accuracy and clarity of discussion of key findings, model choice, diagnostics/remedies, recommendation, and methodological defense.

11. Rubric Performance Standards

Performance Level	General Standard
Excellent	Accurate, complete, well-justified, and financially insightful analysis; strong methodological judgment; results are reproducible and professionally communicated.
Good	Sound analysis with only minor omissions or weaknesses; methods and interpretation are generally appropriate.
Developing	Basic analysis is present but important elements are incomplete, weakly justified, or insufficiently interpreted.
Poor	Major analytical, methodological, interpretive, or reproducibility problems; required components are missing or inappropriate.

12. Generative AI and Academic Integrity

This assessment follows the course generative-AI policy in the syllabus. For group projects, generative AI may be used for refining existing writing, but it may not be used for direct generation of the report. Students remain responsible for the analysis, calculations, interpretation, and originality of the submitted work. Any use of AI must remain within the university and course policy.

13. Final Submission Checklist

☐ The financial question is clear and answerable using the selected regression methods.

☐ Variables and data preparation are documented.

☐ The initial multiple regression is correctly estimated and interpreted.

☐ Statistical inference and model fit are addressed.

☐ Relevant regression diagnostics are conducted.

☐ Problems are addressed with justified remedies where needed.

☐ At least one appropriate advanced regression method is applied and justified.

☐ Financial interpretation is explicit and not merely statistical.

☐ The recommendation follows from the empirical evidence.

☐ Tables and figures are generated through R.

☐ The R Markdown document knits successfully.

☐ The group recording clearly discusses and defends the final regression output.

☐ The submitted files are sufficient to reproduce the analysis.

"The goal is not to find the most complicated model. The goal is to make the best-supported financial decision from the evidence."